

CPCS-203 MID 202

Q1/ multiple choices

1. String s1 =new String("Hi");
 String s2= new String("Bye");
 System.out.println(s1.compareTo(s2));

A) false B) negative number C) 0 D) positive number

2.

String s1 = new String("Chat Gpt");
String s2 = "Chat Gpt";
System.out.print(s1 == s2);
System.out.println(" " + s1. equals(s2));

A) false false B) true true C) true false D) false true

3. String c1 ="C,L;O".replaceAll(",;" , "#") + " ");

A) C L O B) C#L#O C) C,L;O D) C,#L;#O

4. What method should we use to read full directory of a file?

A) getName() B) getPath() C) getAbsolutePath() D) getCanonicalPath()

5. Rabbit [] rabbit = new Rabbit; what is the right way to invoke the method eat() by this array of Rabbit ?

A) Rabbit.eat() B) rabbit.eat() C) rabbit[0].eat() D) this.eat()

6. The output for

System.out.println("c:\\Users\\pc\\eclipseworkspace\\arna\\program.txt".getName());

- a) c:\\Users\\program.txt
- b) c:\\Users
- c) c:\\program.txt
- d) program.txt

7. How to get the length of one dimensional array called student?

- a) `System.out.println("the length is: " + student[i].length);`
- b) `System.out.println("the length is: " + student.length);`
- c) `System.out.println("the length is: " + student[i].length());`
- d) `System.out.println("the length is: " + student.length());`

8. chose the correct output:

```
String [] s = "abc,abc?abc.abc abc#".split("[, ?.]");  
  
for(int i=0; i<s.length; i++)  
    System.out.print(s[i] + " ");
```

- A) abc abc abc abc abc
- B) abc,abc abc.abcabc#
- C) abc abc abc abcabc
- D) abc abc abc abc abc#

9. `StringBuilder x = new StringBuilder ("Hello World");`
`System.out.println(x.insert(5,"HTML and Java"));`

- A) HelloHTML and Java World
- B) HellHTML and Java o World
- C) HelloHTML and Java World
- D) HelloHTML and JavaWorld

Q2/ output

1. explain what happened in writing or drawing:

```
Vehicle V1 = new Vehicle(50);  
Vehicle V2 = V1;
```

2. Write the output:

```
1. public static void main( String[]args){  
String s = "Spring";  
StringBuilder sb = new StringBuilder(s);  
m1(s);  
sb.append(" 2023");  
System.out.println(s);  
System.out.println(sb);  
}  
  
public static void m1(String s) {  
s = s + " 2023";  
}  
  

```

2.

```
public class A{  
private String name;  
private boolean attendance;  
private int ID;  
  
public static void main(String[]args){  
A a=new A();  
System.out.println(a.name);  
System.out.println(a. attendance);  
System.out.println(a.ID);  
}  
  
} //end of class A
```

1. Create a file called midTerm.txt and write the details as shown in the figure. (The file doesn't exist):

--

midTerm.txt
Amal Abdullah 20

2. Read from the previous file (text.txt) and print the details first name, last name, score. Then show the output on the console (The file exist):

Output: First name: Amal Last name: Abdullah Score: 20

Q4/ Complete the following class according to the instructions, and at the end answer the question about whether the class is immutable or mutable:

```
public class Employee {  
    _____ name;  
    _____ id;  
    _____ department;  
    private static int count; //to count employees' number  
  
    public Employee() {  
        //complete the method  
        //invoke 2 arguments constructor  
        //note that the default values for name and id are "unknown" and 0  
  
        _____  
    }  
  
    public Employee (String name, int id) {  
        //complete the method  
        //invoke full arguments constructor  
        //note that department default value is "general"  
  
        _____  
    }  
  
    public Employee (String name, int id, String department) {  
        //Initialize data fields  
  
        _____  
        _____  
        _____  
  
        count++;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public void setName(String name) {  
        this.name=name;  
    }  
  
    public int getId() {  
        return id;  
    }  
  
    public _____ setId(int id) {  
        this.id=id;  
    }  
    public _____ getCount() {  
        _____  
    }  
    //complete the method  
    public _____ EmployeeInfo() {  
        _____  
        _____  
    }  
} //end of class
```

Question: is the previous class mutable or immutable? And explain why: